Chapter 23

The grammatical realisation of apprehensional functions in Ngumpin-Yapa languages (Australia)

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This paper provides a survey of apprehension constructions in the Ngumpin-Yapa (Pama-Nyungan) languages of northern Australia. As in many Australian languages, there are morphosyntactically distinct strategies for encoding apprehension in Ngumpin-Yapa languages: a verbal *apprehensional* morpheme and a nominal *aversive* morpheme. Languages differ in terms of the extent to which these morphemes uniquely encode apprehension, some also encoding physical location or potentiality. We survey apprehensional morphemes in nine Ngumpin-Yapa languages: Walmajarri, Ngardi, Jaru, Wanyjirra, Gurindji, Bilinarra, Mudburra, Warlmanpa and Warlpiri. In these Ngumpin-Yapa languages, the verbal apprehensional morphemes are found in the auxiliary complex as a base for bound pronominals. The nominal aversive morphemes are oblique case suffixes that relate nominals and nominalised predicates to main predicates. The auxiliary bases form part of the system of finite tense–aspect–modality, whereas the case suffixes act as complementisers for non-finite clauses, with dependent tense features. Both classes of morphemes can appear in independent and dependent syntactic environments. Additionally, both classes of apprehensional morphemes can have non-apprehensional functions, for example in Wanyjirra the comitative case suffix introduces concomitant participants, which can also be used to express aversion to particular participants. This chapter makes a substantive contribution to our understanding of apprehension in Australian languages – a topic that hitherto has received little attention, despite the prevalence of apprehensional morphemes in Australian languages. This survey highlights the fact that even in a group of closely related languages, the grammatical encoding of undesirability can vary greatly.

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1 Introduction

In this chapter we survey and compare apprehensional constructions found in languages of the Ngumpin-Yapa subgroup of the Pama-Nyungan family (Australia). We define APPREHENSION as referring to undesirable and avoidable eventualities or entities (since entities with apprehensional marking are employed metonymically for undesirable eventualities, we use the term OUTCOME to encompass both eventualities and entities). Apprehensional constructions may simply comprise an expression of an avoidable and undesirable possible outcome; or may additionally express an alternate course of events which avoids the outcome. This is schematised in Figure 1.

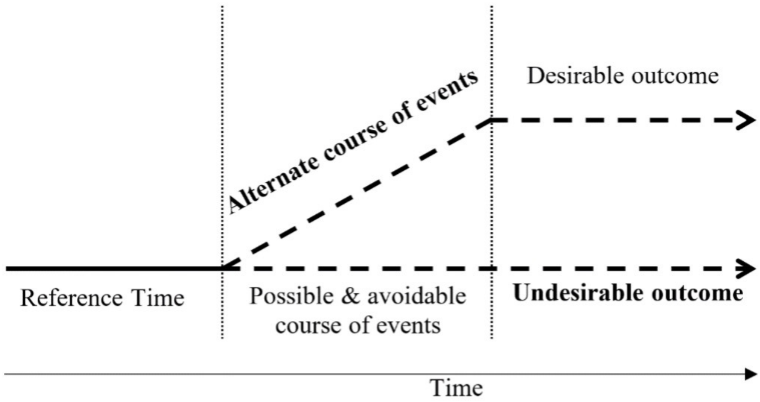

Figure 1: Schema of prototypical meaning of apprehension, where the bolded situations are those which may be overtly expressed in apprehensional constructions.

By UNDESIRABLE, we refer to some event or entity that is evaluated as something that should be avoided. This evaluation is from the speaker's perspective on behalf of the person they are warning, and broadly relates to a negative outcome, whether due to an event, an event associated with an entity, or simply the coincidence of the subject and another entity. By AVOIDABLE we mean that an alternate course of action should be salient to the addressee, as in (1), where the speaker overtly specifies the available course of action through an imperative clause 'get down from there'.

(1) Mudburra (Ngumpin-Yapa)
Karu jurd wandi, bi=ngku linymurr ka-yinarra yali
child get_down fall[IMP] APPR=2SG.NS break be-SPEC that
karndi=ma.
tree=TOP
'Kid, get down from there [alternate course of events], (or else) that tree might break on you [undesirable outcome]!'
[SD:DOS1-2017_017-02:2674149_2677841]

The alternate course of events allowing the undesirable outcome to be avoided may be explicit, as above in (1) or implicit, as in (2). We will refer to a clause which makes the alternate course of events explicit as PREEMPTIVE clause, following Evans (1995: 265). However, the syntactic relation between the two clauses in Ngumpin-Yapa languages is unclear (this is taken up in Section 2.1.3).

(2) Warlmanpa (Ngumpin-Yapa)
Nga=nyanu-n=nga kuma-nma!
APPR=REFL-2SG.S=APPR cut-POT
'You might cut yourself!' [H_K01-505B, DG, 38:14]

Lichtenberk makes a distinction between situations in which the eventuality may be avoided altogether, and where the effects of the eventuality may be mitigated (Lichtenberk 1995: 297), though we have no clear data for the latter in these languages. As will be shown in the survey (Section 1.1), only one Australian language (Martuthunira) is known to grammatically distinguish these two functions. For the purposes of this paper, the alternate course of events may either avoid the event altogether, or mitigate the effects of the event (thus avoiding the undesirable outcome regardless).

Australian languages have a number of lexical and morphosyntactic strategies to convey apprehensional situations. Grammaticalised morphemes with apprehensional functions have been identified within verbal and nominal paradigms (i.e. morphological strategies); we also find grammaticalised multi-clausal constructions which convey apprehension (i.e. syntactic strategies).

In this chapter, we focus on investigating the degree to which apprehension has been grammaticalised in Ngumpin-Yapa languages, a subgroup of the Pama-Nyungan language family. We begin by first surveying apprehension cross-linguistically within Australia (Section 1.1) and present a grammatical overview of Ngumpin-Yapa languages (Section 1.2), leading into a systematic account of apprehensional constructions in these languages. We discuss the various functions of the aversive case (relational, subordinating, and insubordinate functions signalling apprehension, Section 2.1); the marking of complements of 'fear' predicates (Section 2.2), as well the encoding of apprehension by a morpheme known as the auxiliary (Section 2.3). Finally, we consider the range of functions that sentences involving apprehensionals tend to serve in discourse, divided into directives and assertives (Section 3).

1.1 Apprehension in Australian languages

In descriptions of Australian languages, apprehensional morphemes have been assigned a number of terminological labels, reflecting the lack of a unified approach to understanding these morphemes. Table 1 exemplifies the wide array of glosses given to apprehensional morphemes in grammars of Australian languages.

Moreover, while many Australian languages appear to have some strategy for encoding apprehensional meanings, there has been very limited literature examining how these constructions vary across the continent.[1] A singular exception is Zester's (2010) survey of apprehensional constructions in 8 Pama-Nyungan languages and 4 non-Pama-Nyungan languages, which sought to compare the category of apprehension in Australian languages and English.[2]

We categorise apprehensional morphemes into three distinct types based on morphosyntactic locus of exponence, described below:

1. nominal case inflections;
2. verbal mood inflections; and
3. non-inflecting particles.

1.1.1 Nominal case suffixes

Apprehensional suffixes in Australian languages occur within nominal paradigms, either as dedicated case suffixes or as a function of a case suffix not primarily apprehensional in nature. In Wargamay (Dyirbalic), for example, a locative suffix has a range of adjunct spatial functions, such as to mark location in space and time ('in', 'at' 'on', 'near' etc.), accompaniment ('with') and circumstance ('while'). The Wargamay locative case can also be used in apprehensional contexts, as in (3), in which the locative suffix *-nga* is used to indicate that the white man is something to be avoided.

[1] Gaby (2017) presented an unpublished typology of apprehension in Australian languages.

[2] The Pama-Nyungan languages included in Zester's study were: Yidiny, Djabugay, Pithanthathara, Martuthunira, Panyjima. The non-Pama-Nyungan languages were Ndjébbana (Arnhem), Gooniyandi (Bunuban), Wambaya (Mirndi) and Jingulu (Mirndi). No Ngumpin-Yapa languages were covered in Zester's survey.

Table 1: Sample of the various glosses given to apprehensional morphemes.

Language	Morpheme(s)	Gloss	Reference
Djabugay	*lan, -ndabi, -ndjabi*	'aversive'	(Patz 1991: 268)
Wangkajunga	*-ngkamarra*	'avoidance'	(Jones 2011: 95, 290–291)
Kayardild	*waalu-tha/-waal-i-ja*	'evitative'	(Evans 1995: 173–175)
Warumungu	*-kajji*	'lest'	(Simpson & Heath 1982: 22–23)
Alyawarra	*-ikitja*	'negative causative'	(Yallop 1977: 69, 75–76)
Walmajarri	*-ngkamarra*	'projected reason'	(Hudson 1978: 31)
Mangarrayi	*baḷaga*	'evocative/anticipatory'	(Merlan 1989: 146–147)
Ngarinyman	*ngaja*	'apprehensive'	(Angelo & Schultze-Berndt 2016)
Bilinarra	*ngaja*	'admonitive'	(Meakins & Nordlinger 2014: 305)

(3) Wargamay (Dyirbalic; Dixon 1981: 30)
Ngayba bimbirigi waybala-**nga**.
1SG.S run.PERF white_man-**LOC**
'I'm running for fear of the white man.'

Other languages possess dedicated nominal suffixes for encoding apprehension. In the Australianist tradition, these are typically labelled 'evitative' or 'aversive' cases.[3] These suffixes encode that the referent of the case-marked phrase is to be avoided, often translated with some implicit eventuality associated with the referent, as in the Yidiɲ example in (4).

(4) Yidiɲ (Yidinic; Dixon 1977: 262)
Jaja munubujun nyinany / bama-**yida**.
child.ABS inside:STILL sit:PST person-**AVE**
'The child still sat inside [the hut] for fear of [being seen by] the people.'

In some languages, the dedicated case suffix is formally comprised of another case suffix as well as an apprehensional increment. In Pintupi, for example, to form the 'avoidance' case, the increment *-marra* is added to a locative case form which marks accompaniment, animate goals, and animate locations (Hansen & Hansen 1978: 57). Examples (5) and (6) demonstrate the same stem *mul̠ipuru* 'one obscured by the shelter' in the locative (*-ngka*) and avoidance cases (*-ngkamarra*) respectively. This type of complex case formation is cross-linguistically common in Australia, and has been termed "compound case" (Schweiger 2000: 256) or "parasitic formation" (Matthews 1991: 85–86).

(5) Pintupi (Western Desert; Hansen & Hansen 1978: 99)
Tjampitjin-ta=litju=lu watjan̠u, mul̠i-**puru-ngka**.
name-LOC=1DU.EXCL.S=3SG.LCT said shelter-**OBSC-LOC**
'We spoke to Tjampitjinpa who was obscured by the shelter.'

(6) Pintupi (Western Desert; Hansen & Hansen 1978: 99)
Mul̠i-**puru-ngkamarra**=n̠a=lura yarka-∅ pitjangu.
shelter-**OBSC-AVE**=1SG.S=3SG.AVE secret-NOM went
'To avoid that one who is obscured by the shelter, I kept myself hidden as I walked.'

[3] Henceforth we will generally refer to isomorphic case forms encoding apprehension as 'aversive' cases and gloss them consistently as 'AVE'.

1.1.2 Verbal inflection

Some Australian languages have a dedicated verbal inflection which signals apprehension. Martuthunira (Ngayarta, Pama-Nyungan), for example, has a 'lest' verb suffix which encodes apprehensional circumstances, exemplified in (7).

(7) Martuthunira (Ngayarda; Dench 1995: 73)
Nhiyu pawulu wangka-yangu puni-layi karla-ngku **kampa-wirri**, mir.ta waruul puni-nguru.
this child tell-PASSP go-FUT fire-EFF **burn-APPR** not still go-PRS
'This child has been told to go lest he gets burnt by the fire, but he still hasn't gone.'

A less common locus of encoding apprehensional meanings across the Australian continent is that of a discontinuous inflection as in Nyangumarta (Marrngu, Pama-Nyungan) (Sharp 2004: 165, 186–187), or a clitic as in another Marrngu language, Mangarla (Agnew 2020: 335) which takes predicates as its hosts, as shown in (8). In the case of the latter, the status of the ANTICipatory in Mangarla as a clitic is affirmed by its co-occurrence with TAM inflections other than the imperative and its (rare) encliticisation to nominal predicates.[4]

(8) Mangarla (Marrngu; Agnew 2020: 336)
Pantu warlu parra~parra kampa=nyjurruny=**kulu**.
MED fire heat~RDP burn.IMP=1PL.INCL.O=**ANTIC**
'That fire is hot and might burn us.'

There is variability across languages with respect to whether a given verbal inflection is a dedicated marker of apprehension or semantically less specific, e.g. whether it also has a general irrealis meaning denoting any possible and unrealised events which need not be negatively evaluated. For instance, Wargamay has an 'irrealis' verbal suffix which can be used in future, unreal and apprehensive contexts. The 'irrealis' inflection is exemplified in (9), where the event is not negatively evaluated, and in (10), where it is.[5]

[4]The verb in (8) is glossed as an imperative, which expresses a wide range of functions in Mangarla (Agnew 2020: 320).

[5]Further examples are: the anticipatory inflection in Ngardi (Ngumpin-Yapa, Pama-Nyungan; Ennever 2018), the subjunctive inflection in Yulparija (Western Desert, Pama-Nyungan; (Burridge 1996: 39) or the obligative and hypothetical inflections in Wangkajunga (Western Desert, Pama-Nyungan; Jones 2011: 201–202). All of these inflections share a semantic basis that the speaker is conveying that the event is likely to occur. Undesirability does not seem to be encoded. In some languages, clausal negation affects possible interpretations, as in Mangarla (Agnew 2020: 335–339).

(9) Wargamay (Dyirbalic; Dixon 1981: 54)

Ngayba	nga:	**walama**.
1SG.S	NEG	**ascend.IRR**

'I'm not climbing (anymore, because I'm tired).'

(10) Wargamay (Dyirbalic; Dixon 1981: 54)

Ngaru	gilwalga /	**ba:dima**.
NEG	kick.NEG.IMP	**cry.IRR**

'Don't kick him, lest he cry / he might cry.'

In most Australian languages, where a verb suffix denotes an apprehensional meaning, the clause in which it occurs is usually either juxtaposed or subordinate to a main clause encoding a preemptive event, as in (11), suggesting these verb inflections may be indicative of subordination.

(11) Yidiɲ (Yidinic; Dixon 1977: 351)

Nyundu	gajidagan /	giyara-nggu	guba:-**nji**.
you:ERG	long_way:INCH:VBLSR:IMP	stinging_tree-ERG	burn-**LEST**

'You keep away, lest you get stung by the stinging tree!'

1.1.3 Non-inflecting particles

Some Australian languages have particles (often complementisers) which introduce a subordinate clause and place it in an apprehensional relation to the (preemptive) main clause. Zester (2010) discusses two non-Pama-Nyungan languages in Australia's north exhibiting apprehensional complementisers: the Mirndi languages Wambaya (Nordlinger 1998: 210) and Jingulu (Pensalfini 2003). An example from Jingulu is given in (12), where the subordinate clause is headed by the evitative complementiser *karningka* (glossed 'LEST').

(12) Jingulu (Mirndi; Pensalfini 2003: 120)

Buba	jimi-rni	ila-mi	jungkali	marru-ngka-mbili	**karningka**
firewood	that(n)-FOC	put-IRR	far	house-ALL-LOC	**LEST**

buba-arndi	ngunji-jiyimi	marru.
fire-INST	burn-come	house

'Put the firewood down far from the house, lest the fire burn the house down.'

1.1.4 No apprehensional morphemes

Other languages may convey apprehension by the juxtaposition of a preemptive clause with a clause expressing a possibility, but without overt apprehensional morphology on the latter. In Warluwarra (Ngarna), there are no morphemes which could be clearly associated with apprehension. Instead, apprehension is derived via pragmatic enrichment in irrealis situations, particularly when juxtaposed with a preemptive clause, as in (13).[6] In this case, if the addressee smells the meat (preemptive), it mitigates the effects of it being rotten (for example, it results in them not eating it if it is rotten).

(13) Warluwarra (Wagaya-Warluwaric; Breen 1970: 234)

Jatjunma	jinja	karlaanha /	warrɣa	ngarlatha	jiwa.
smell.IMP	that[ACC]	meat[ACC]	rotten	maybe	it

‘Smell that meat; it might be rotten.’

Ngumpin-Yapa languages present an opportunity to refine cross-linguistic generalisations regarding apprehensional constructions, particularly in the Australian context. Firstly, most Ngumpin-Yapa languages exhibit both a nominal case suffix and a particle-like auxiliary to signal apprehension, unlike any other language surveyed in Australia. Additionally, Ngumpin-Yapa languages, despite being a close-knit group of languages, show a very interesting diversity in the expression of complements of ‘fear’ predicates and apprehensional marking in general.

1.2 Ngumpin-Yapa languages

This survey focuses on nine languages of the Ngumpin-Yapa subgroup, a subgroup within the widespread Pama-Nyungan family in Australia. The languages for which data is discussed in this paper are: Walmajarri, Jaru, Ngardi, Gurindji, Wanyjirra, Bilinarra, Mudburra, Warlmanpa and Warlpiri. Our current understanding of their relative cladistic relationships within the Ngumpin-Yapa subgroup is represented in Figure 2. The areal distribution of Ngumpin-Yapa languages across northwestern Australia is shown in Figure 3.

Evidence for the unity of the Ngumpin and Yapa subgroups, and for their position within the Nyungic subgroup, is provided by Laughren & McConvell

[6]Note that this is similar to languages which have an irrealis inflection or complementiser-like particles which are used in apprehensional constructions but do not encode specifically ‘apprehension’. In fact there may be no distinction between these two types. The distinction lies in how strongly the morphemes are associated with apprehension.

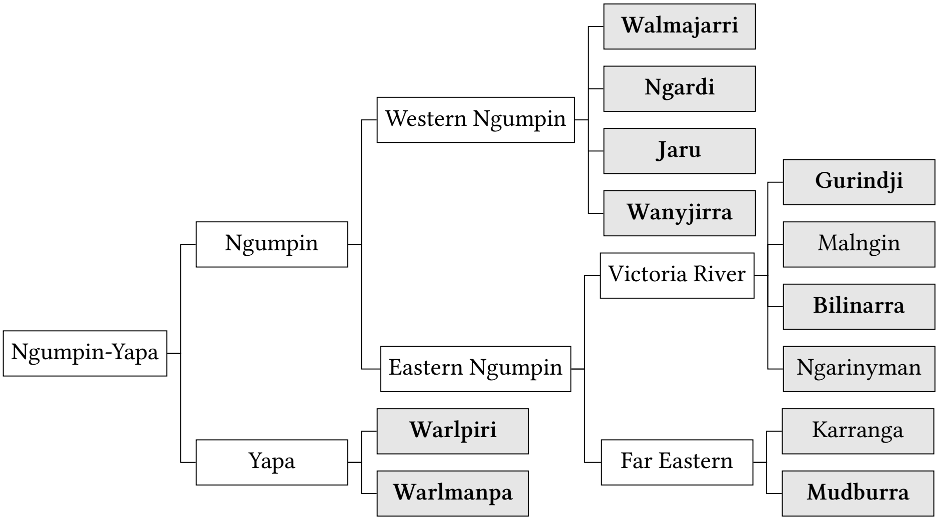

Figure 2: Phylogeny of the Ngumpin-Yapa subgroup. Languages included in this study are emphasised. Adapted from McConvell (2009: 791)

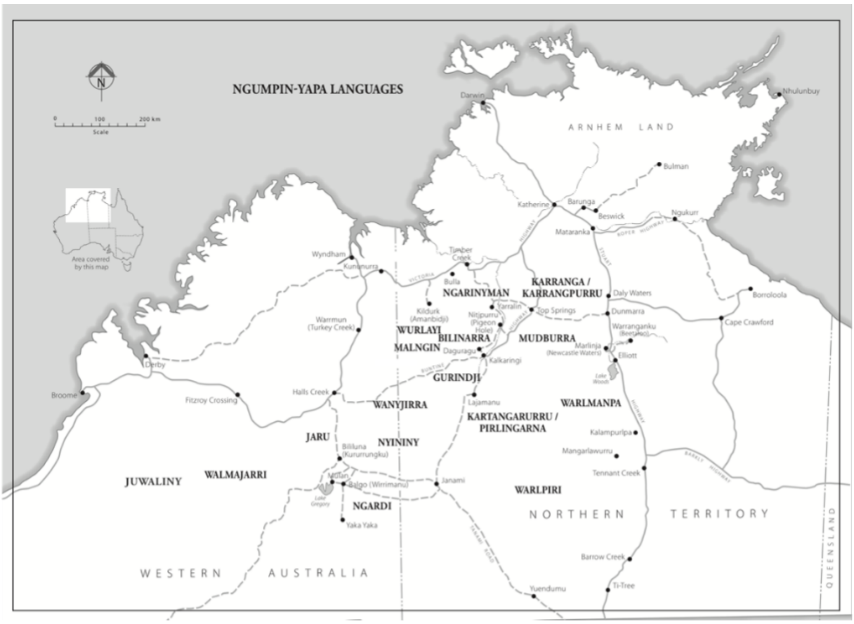

Figure 3: Map of relative locations of Ngumpin-Yapa languages across northern Australia (adapted from Meakins & Nordlinger 2017; cartographer: Brenda Thornley).

(2004), while evidence adduced from pylogenetic studies is provided by Bowern & Atkinson (2012) on the basis of lexical data, and by Macklin-Cordes & Round (2015) on the basis of phonotactic data. In this section, we briefly detail some of the relevant morphosyntactic characteristics of these languages.

A working orthography of all the languages used in this survey is presented in Table 2. The only phoneme not shared across the subgroup is a retroflex flap which is only fully phonemicised in certain dialects of Warlpiri and is otherwise an allophone of the retroflex stop (Western Warlpiri) or an allophone of the retroflex continuant (glide) phoneme (Ngardi and Jaru). As for stops, voicing is not distinctive in the languages under discussion. Across the Ngumpin-Yapa languages there are three orthographies in use which largely differ by the representation of stops.[7] We retain the orthography from source materials in examples.

Table 2: Maximal Ngumpin-Yapa consonant phoneme inventory (orthographic forms given in brackets where they differ from the expected IPA symbol)

	Peripheral		Apical		Laminal
	Bilabial	Velar	Alveolar	Retroflex	Palatal
Stop	p/b	k/g	t/d	ʈ <rt/rd>	c <j>
Nasal	m	ŋ <ng>	n	ɳ <rn>	ɲ <ny>
Lateral			l	ɭ <rl>	ʎ <ly>
Flap/Trill			r <rr>	ɽ <rd>	
Glide	w			ɻ <r>	j <y>

All Ngumpin-Yapa languages have a three-way distinction in vowel quality with most languages generally described as having a marginal length contrast with a low functional load (Table 3).

Table 3: Ngumpin-Yapa vowel inventory

	Front	Back
High	i, (ii)	u, (uu)
Low	a, (aa)	

[7]p ~t ~k: Gurindji, Warlpiri, Ngardi, Warlmanpa
b ~d ~k: Mudburra
b ~d ~g: Bilinarra, Ngarinyman, Wanyjirra

Ngumpin-Yapa languages are agglutinating and purely suffixing. Core arguments are case-marked and are also cross-referenced by bound pronominal enclitics which encode person, number and case features. These languages all demonstrate the properties of non-configurationality as described for Warlpiri (Hale 1983), including: free word order, discontinuous phrase structure and frequent ellipsis of nominals. All Ngumpin-Yapa languages share the property that the pronominal clitics generally occupy second position of the clause, so-called Wackernagel's position (Wackernagel 1892). Other features of these clitics vary considerably across the subgroup – both with respect to the range of grammatical relations that bound pronouns may cross-reference, and the types of clausal elements to which they may attach.

One type of host for the pronominal clitics are termed *auxiliary bases.* These are otherwise uninflecting particles which obligatorily host the pronominal clitics. The semantics of the auxiliary base varies considerably across languages. In some cases, the auxiliary base is essentially a meaningless host for the pronominal enclitics and has been termed "catalyst" by some authors, for example *ba* in Mudburra (Capell 1956: 68–69), *ngu* in Gurindji (McConvell 1996b), and *nga* in Ngarinyman (Jones et al. 2019: 30). In some Ngumpin-Yapa languages, there is a small inventory of semantically meaningful auxiliaries (or "modal particles"), which contribute tense, aspect, or mood information to the clause. For example, the auxiliaries in Warlpiri include *lpa* 'imperfective' and *ka* 'present' (Laughren 2013), which combine with verb inflections to encode fine-grained TAM distinctions.[8]

In nearly all Ngumpin-Yapa languages, lexemes of certain other grammatical categories also attract the pronominal enclitics (in lieu of an auxiliary base in the given construction), including complementisers, interrogatives as well as imperative-inflected verbs. In languages without a catalyst auxiliary, such as Ngardi, in the absence of any of these more typical hosts, the bound pronouns encliticise to the first phonological word (or in some cases, the first phrase) (Ennever 2021: 310–313).

These bound pronominals pattern primarily according to a nominative-accusative system, in which A and S are encoded by a subject series, and O is encoded by an object series. For many of the languages surveyed, there is syncretism between the object (O) and oblique (OBL) series in all subject/number combinations except 3SG, leading to the convention of labelling these forms as either O 'object/oblique', or NS 'non-subject'. In nearly all of the Ngumpin languages, nominals in certain other (traditionally semantic) cases such as the

[8] Western Warlpiri has a future verb inflection in addition to the auxiliaries.

locative, allative and ablative can also be cross-referenced by bound pronominals (which, if distinct from the oblique, are labelled LCT 'locational' here), whereas this is not permitted in either of the Yapa languages. Language-specific systems are given in Table 4. Ngumpin-Yapa languages do not utilise the bound pronominal system for aversive case-marked adjuncts (Section 2.1), but do for complements of 'fear' predicates (Section 2.2).

Table 4: Relationship between case marking and the cross-referencing bound pronouns.

		Subject			Non-subject			
	Case	ERG	ABS	ABS	DAT	LOC	ALL	ABL[a]
Yapa	Warlpiri	S	S	O	OBL	—	—	—
	Warlmanpa	S	S	O	OBL	—	—	—
Eastern Ngumpin	Bilinarra	S	S	O	OBL	—	—	—
	Gurindji[b]	S	S	O	OBL	LCT	LCT	LCT
	Mudburra	S	S	O	OBL	OBL	OBL	—
	Wanyjirra	S	S	O	OBL	OBL	OBL	OBL
Western Ngumpin	Jaru	S	S	O	OBL	LCT	LCT	LCT
	Ngardi	S	S	O	OBL	LCT	LCT	—
	Walmajarri	S	S	O	OBL	ACS	—	—

[a]We use the case label *ablative* here, however *elative* is used for some languages.
[b]This follows McConvell's (1996b: 84) analysis of bound pronouns in Gurindji, where "affected locations … are cross-referenced by clitics, unlike other locations, and in a way distinct from either S, O, or IO clitics."

All Ngumpin-Yapa languages exhibit extensive case marking. Nominals bear no overt case marking in the S and O function, and bear overt ergative case marking in the A function (in accordance with an ergative-absolutive case-marking pattern). While free pronouns in Warlpiri, Ngardi, Jaru, Walmajarri, and Eastern Mudburra exhibit ergative-absolutive marking like other nominals, in Wanyjirra, Gurindji, Bilinarra, Western Mudburra and Warlmanpa, free pronouns are invariably unmarked in all three core functions.

All Ngumpin-Yapa languages possess two morphosyntactically distinct parts of speech with verbal semantics: inflecting verbs and coverbs. Inflecting verbs are a closed class of inflecting words (from around thirty to over a hundred) which bear tense/aspect/mood inflection; however the languages vary considerably in terms of the number and type of features encoded by verbal inflections. Variation is found, for example, in the distinctions made for past time reference (with a particularly differentiated system in Jaru and Ngardi), the presence of associated motion markers (Mudburra, Ngardi, Warlpiri, Warlmanpa), the presence of directional suffixes which can be cumulatively realised with another feature (Ngardi, Warlpiri), and patterns of reduplication (with a particularly complex system in Mudburra). An inflecting verb may constitute a simple predicate on its own, or it may act as a light verb in a complex predicate along with a coverb (also called a "preverb"; see Osgarby & Bowern 2023).

Ngumpin-Yapa languages exhibit both finite and non-finite subordination. The latter typically involves a non-finite verb form or a coverb bearing complementiser case suffixes, expressing the relative time and, in some cases, control properties of the subordinate clause (Simpson & Bresnan 1983). Finite subordination is achieved by the use of a complementiser particle within the finite subordinate clause. Such clauses may have both temporal (T-relative) or anaphoric (NP-relative) functions; however multiple interpretations for the relationship between clauses are usually possible (Hale 1976).

2 Morphosyntax of apprehension in Ngumpin-Yapa languages

The complete inventory of grammatical markers of apprehension in Ngumpin-Yapa languages is presented in Table 4 and visually represented in Figure 4. Three loci of grammaticalised apprehensional marking have been identified. The first is what will be termed an *aversive case*, a dedicated nominal case suffix, which can mark a nominal referent as something which is feared, or a non-finite clause which denotes an undesired outcome. The second is the idiosyncratic marking of complements of 'fear' predicates. These are treated as a grammaticalised form of apprehensional marking since their marking is idiosyncratic but shows some relationship to the grammar of apprehension elsewhere in the languages. The third is an auxiliary (although in some languages these auxiliaries are more complementiser- or particle-like) which indicates that the event referred to by the finite clause is undesired. While most languages have all three constructions available, Bilinarra and Jaru lack an aversive case suffix.

The aversive case is discussed in Section 2.1, first exemplifying its relational function (Section 2.1.1), then its subordinate function (Section 2.1.2), before discussing the issues in accurately distinguishing the two functions (Section 2.1.3).

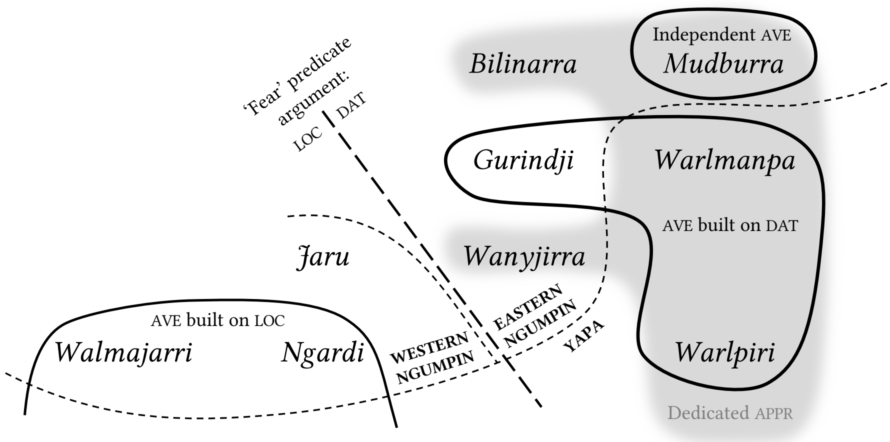

Figure 4: Geographical distribution of dedicated apprehensional markers in Ngumpin-Yapa languages.

Furthermore, in some languages these cases exhibit insubordinate functions (Section 2.1.4). 'Fear' complementation is discussed in Section 2.2: strategies include the locative case (Section 2.2.1), the dative case (Section 2.2.2), as well as the aversive (Section 2.2.3). These variable case-marking patterns are summarised in Table 5. Finally, in Section 2.3 we analyse the auxiliaries, discussing their syntactic properties in Section 2.3.1 and whether they are dedicated to encoding apprehension or not in Section 2.3.2.

2.1 Aversive case

Six Ngumpin-Yapa languages possess an aversive case suffix that can be classified as a dedicated apprehensional morpheme. Three languages lack a dedicated case suffix for expressing apprehension, and instead allow apprehensional meanings to be encoded by a polyfunctional case such as locative or comitative. In Figure 4 we distinguish between aversive case suffixes that are formally independent of other suffixes (Mudburra), those built on the dative case suffix (Gurindji, Warlpiri, Warlmanpa), and those built on the locative case suffix (Walmajarri, Ngardi). Warlpiri, Warlmanpa (Yapa) and Gurindji (Eastern Ngumpin) all have aversive case forms which seem to be comprised of the dative case suffix *-ku* ~*-wu* plus an increment, together marking the aversive case.

Table 5: Overview of morphemes used to signal apprehension in Ngumpin-Yapa languages.

	Language	Case in aversive function	Case on 'fear' complement	Auxiliary[a]
Yapa	Warlpiri	*-kujaku* AVE	DAT, AVE	*kalaka/kajika* POT
	Warlmanpa	*-kuma* AVE	DAT	*kala(ka)* APPR *nga=…=nga* FUT=…=POT
Eastern Ngumpin	Bilinarra	*-ngka, -la, -Ca* LOC	DAT	*ngaja* ADM
	Gurindji	*-kumpalng*, *-wumpalng* AVE	DAT	*ngaja* ADM
	Mudburra	*-wirri* AVE	DAT, AVE (rare)	*bi(ya)* APPR
	Wanyjirra	*-kunyja*, *-wunyja* COM	DAT	*ngarra* ADM
Western Ngumpin	Jaru	*-ngka, -la, -Ca* LOC	LOC	*ngarra* 'possible'
	Ngardi	*-Camarra* AVE	LOC, AVE	*ngarda* HYP
	Walmajarri	*-Camarra*, *-karrarla* AVE	LOC, AVE (rare)	*nga-rta* MR1-DUB *pa-rta* MR2-DUB

[a]Note that in the *auxiliary* column we have retained the authors' glossing, however elsewhere we have glossed these auxiliaries APPR in order to allow ready comparison across languages. The *-Ca* forms in the *aversive function* column represent forms which exhibit extensive allomorphy conditioned by the preceding segment and the syllable or mora count of the stem.

Two Western Ngumpin languages (Walmajarri and Ngardi) possess aversive suffixes formally built on a locative case allomorph *-Ca* + an increment *-marra*. This case formation (LOC + *marra*) is widespread in the Western Desert languages to the south of Walmajarri and Ngardi, for example Kukatja and Wangkajunga. A further variant is *-karrarla* in Walmajarri. Interestingly, a locative formative *-rla* is still involved in *-karrarla* but as the right-most constituent, with a base *-karra*. An example is provided in (14).

(14) Walmajarri (Western Ngumpin; Richards & Hudson 2012)

Kiwiri	ma=rnalu	jart-kuji-lany	mimi-**karrarla.**
wood	MR1=1PL.EXCL.S	burn-CAUS-CUST	sick-**AVE**

'We would burn the *kiwiri* wood to keep away sickness.'

Like many other case suffixes in Ngumpin-Yapa languages, the aversive case can either mark a participant in a particular semantic relation to the clausal predicate (Section 2.1.1) or it may serve as a subordinating case suffix (Section 2.1.2). These functions have been termed *relational* and *complementiser* functions by Dench & Evans (1988) – although the two are not always easily distinguished (Section 2.1.3). Additionally, aversive cases may appear in insubordinate constructions (Section 2.1.4). These constructions lack a main clause predicate and instead the implied event associated with the participant marked by the case is insubordinated, i.e. appears in a main clause function (see Evans & Watanabe 2016). In all cases in the corpora that we are aware of, the aversive case is used to indicate an entity or situation which is to be avoided, rather than indicating a situation to be mitigated.

2.1.1 Relational function

Aversive case-marked NPs in all main clause types introduce a referent to be feared/negatively evaluated; and prototypically provide a justification for the action described by the remainder of the clause. Examples include (15) to (17).

(15) Mudburra (Eastern Ngumpin)[9]

Yurrub	wandi	**ngarrambalyangka-wirri**.
hide	fall.IMP	**police-AVE**

'Hide, for fear of the police!'[10] [MSD: AHA1-2016_027-03: 1708833_1711002]

[9]Mudburra, Ngardi, and Warlmanpa data come directly from corpora. The reference format of Mudburra and Ngardi is: Speaker initials: Audio code: Timestamp. The reference format of the Warlmanpa data is: Audio code: Speaker initials: Timestamp.

[10]In Ngumpin-Yapa languages (and indeed other Australian languages with complex predicates), there is not always a straightforward link between the glosses of predicates and the given trans-

(16) Ngardi (Western Ngumpin)
Wakurra=rna wirriri ya-ni **mula-ngkamarra ngantany-jamarra.**
NEG=1SG.S back go-PST **this-AVE man-AVE**
'I didn't go back for fear of this man.' [PSM: TEN1-2017_008-02: 938110_946191]

(17) Walmajarri (Western Ngumpin; Richards & Hudson 2012)
Jilngin-tamarra ma=rna kaniny~kaniny yuka-lku.
dew-AVE MR1=1SG.S inside~RDP lie-POT
'I'll sleep inside on account of the dew.'

In some languages, such as Jaru (Western Ngumpin), the locative case can encode apprehension independent of a 'fear' predicate as in (18). Further, in Warlpiri and Bilinarra, the locative-marked nominal can comprise the entire utterance, as in (19)–(20). Similarly, the comitative case (which prototypically encodes accompaniment) has an additional apprehensional function in Wanyjirra, as in (21).

(18) Jaru (Western Ngumpin; Tsunoda 1981: 58)
Wanyja-rra gardiya-**la**!
leave-IMP white_man-**LOC**
'Leave (it) for fear of the white man (if you touch it, he might get angry).'

(19) Bilinarra (Eastern Ngumpin; Meakins & Nordlinger 2014: 129)
Gawayi! Gururrij-**ja**!
come_here car-**LOC**
'Come here! Look out—car!'

(20) Warlpiri (Yapa; Mary Laughren, p.c.)
Warlu-**ngka**!
fire-**LOC**
'Look out for the fire!'

(21) Wanyjirra (Eastern Ngumpin; Senge 2015: 200)
Ngu=rna=yanunggula wulaj garriny-ana yalu-**wunyja** garaj-**guja**
REAL=1MIN.S=3AUG.LCT hide stay-PRS DIST2-**COM** human-**COM**

lation, as is the case in this example where the inflecting verb is glossed 'fall' yet this does not form part of the translation. It is often the case that the coverb bears the lexical information, and the inflecting verb essentially only bears grammatical (tense, mood, and aspect) information.

ngaringga-**wuja**.
woman-**COM**
'I hide from that man and woman.'

Aversive-marked NPs generally only have adjunct status in all Ngumpin-Yapa languages, and as such they are never cross-referenced in the bound pronoun.[11] This will be contrasted with complements of 'fear' predicates in the dative and locative cases which are treated as clausal arguments (Section 2.2).

2.1.2 Subordinating function

Aversive cases can also mark non-finite subordinate 'lest' clauses, as in (22).

(22) Warlmanpa (Yapa)
Pingka pa-nka **wa-nja-kuma**.
slowly go-IMP **fall-INF-AVE**
'Go slowly lest you fall.' [H_K01-505B: DG, 43:43]

However, they are not necessarily dedicated apprehensional markers, and may more broadly mark undesirable events (23), or events without a clear preemptive action (24); or in the case of Warlpiri, even impossible (yet desired) events (25).

(23) Warlpiri (Yapa; Laughren 2015: 53)
Pirtiri-ma-ni ka **ya-ninja-kujaku** janjanypa-rlu.
pester-CAUS-NPST PRS **go-INF-AVE** demanding-ERG
'The demanding (child) is pestering her not to go.'

(24) Mudburra (Eastern Ngumpin)
Wirlarnkarra ba=rla karri-nyarra **barany-karra-wirri**, jirrbu
frightened CAT=3SG.DAT be-NARR **slip-D.LOC-AVE** drown
wandi-nya-wirri.
fall-INF-AVE
'He was scared of it (the water), for fear of slipping, for fear of drowning.'
[MAC: PMC1-M9-01: 2684588-2688547]

(25) Warlpiri (Yapa; Laughren 2015: 6)
Wiri ka=rna nyina-mi yalumpu=ju pirnki-kirra **yuka-nja-kujaku**.
big PRS=1SG.S sit-NPST that=TOP cave-ALL **enter-INF-AVE**
'I'm too big to enter the cave.'

[11] See the Walmajarri example in Section 2.2.3 as an isolated exception.

There are no clear constraints on argument co-reference between main and non-finite subordinate 'lest' clauses in any of the Ngumpin-Yapa languages examined here. The subject of the subordinate aversive clause can be co-referent with the subject of the main clause as in (24). Conversely, there can be co-reference between the object of the main clause and the subject of the subordinate clause, as in (26).

(26) Warlpiri (Yapa; Laughren 2015: 5)
[Kurdu-kujaku pi-nja-kujaku]=rna yirra-rnu.
[child-AVE bite-INF-AVE]=1SG.S place-PST
'Lest (it) bite the child I put it (the dog) away.'

2.1.3 Main clause or subordinate clause?

The previous two sections considered relational functions of aversive case (i.e. a nominal modifier to a main clause) and subordinate functions of aversive case (i.e. subordinate clauses related to the main clause by an apprehensional relation) as clearly discrete constructions. However, in Ngumpin-Yapa languages this distinction is not always straightforward.

The close formal similarity between the relational and subordinate usage of the aversive case in these languages can be exemplified with the Warlpiri data below. In (27a), there is a subordinate clause, headed by a non-finite verb marked with aversive case, and a nominal which is optionally marked with aversive case. However, either of these constituents may be elided: in (27b), the nominal does not surface, although this is still likely to be considered a subordinate clause as the non-finite verb surfaces. In (27c), the verb is elided and only the nominal surfaces, which is formally undifferentiated from any other nominal modifier, and the event associated with the nominal *karli* 'boomerang' is inferred. Thus there is no clear separation between a subordinate clause usage of the aversive case and a relational usage.

(27) Warlpiri (Yapa; Laughren 2015: 2)

a. Yapa=ka wuruly-parnka karli(-kijaku) luwa-rninja-kijaku.
man=PRS hide-run.NPST boomerang-AVE strike-INF-AVE
'Someone's running away to not get struck by a boomerang.'

b. Yapa=ka wuruly-parnka luwa-rninja-kijaku.
man=PRS hide-run.NPST strike-INF-AVE
'Someone's running away to not be struck.'

c. Yapa=ka wuruly-parnka karli-kijaku.
man=PRS hide-run.NPST boomerang-AVE
'Someone's running away to avoid (being struck by) a boomerang.'

In (27c), despite there being no overt verbal predicate providing evidence for a non-finite subordinate clause, the aversive case here can be viewed as functioning as a dynamic predicate (Laughren 2015). The blurring of clear distinctions between relational and subordinating usages of case is something of a feature of all Ngumpin-Yapa languages and is captured neatly by Hale's (1982: 285) usage of the term "vague predicational use of case" to refer to case marking of a nominal which implies the presence of a verbal predicate. Laughren (2015) goes as far as to argue that aversive *-kujaku* always has scope over an event rather than solely a participant of a clause.

2.1.4 Insubordinate function

Finally, there are some select, additional uses of case marking in apprehensional contexts in what can be categorised as insubordinate usages of case (see Evans 2007, Evans & Watanabe 2016), that is, the use of case as a fully independent predicate. An example of this was provided in (19) involving the polyfunctional locative case in Gurindji. To this we can add independent usages of the aversive case in Mudburra. Insubordinate constructions of this type are found with both nominal (28) and non-finite verbal (29) forms. Both utterances are used with the illocutionary force of a directive and coerce the addressee to some action, which corresponds to the implied main clause in these uses of the aversive.

(28) Mudburra (Eastern Ngumpin)
Kardajala-wirri!
sandhill_people-AVE
'Careful of the sandhill people!' [MLD: RGR1-T27A-01: 819656_8211120]

(29) Mudburra (Eastern Ngumpin)
Janki-nyi-wirri!
burn-INF-AVE
'Careful, you might get burnt!' [MLD: RGR1-T56A-01: 836058_836794]

2.2 Complements of 'fear' predicates

In the preceding section, a type of apprehensional construction was described in which a case suffix introduces a semantic relation between an NP or an adverbial clause and the main clausal predicate. In all of the examples provided, the aversive cases mark syntactic adjuncts and are licensed by their own semantic relation which they establish between their host and the clausal predicate (or, in the case of the insubordinate function, an implied predicate).

Apprehensional meanings can also be conveyed by predicates encoding fear, for example 'to be afraid'; 'to be frightened of' and so forth. 'Fear' predicates in Ngumpin-Yapa languages all select intransitive subjects but exhibit interesting variation with respect to how the complement (the SOURCE/OBJECT OF FEAR) is grammatically encoded. This variation can be observed in the choice of case marking but also in how these arguments are (or are not) cross-referenced in the pronominal complex.

In Ngumpin-Yapa languages, 'fear' predicates are only found as complex verbs. The lexical element expressing a state of fear is either a nominal (e.g. Walmajarri and Ngardi) or a coverb (e.g. Gurindji), in either case combining with an inflecting verb which encodes TAM information. To illustrate briefly with an example from Ngardi, the nominal *rayin* 'afraid' combines with a generic stative verb (e.g. *karri-* 'be') which is inflected for TAM. The resulting argument structure of this particular complex predicate (*rayin=karri-*) is that of an extended intransitive involving an absolutive subject and a locative case-marked "object" denoting the source of fear. Both the absolutive subject and the locative "object" are also cross-referenced in the enclitic bound pronouns which index subject, person and argument features.

Predicates of 'fear' in Ngumpin-Yapa languages may select an argument in a range of seemingly idiosyncratic case forms, including the locative (Section 2.2.1), dative (Section 2.2.2), and aversive (Section 2.2.3). As shown in Table 6, Ngumpin-Yapa languages use either the locative or the dative, and some languages allow the aversive case instead of the locative or dative.

2.2.1 Locative

Those Ngumpin-Yapa languages which can mark semantic sources of fear with the locative case can also use the locative case relationally to link a complement to a 'fear' predicate. This functionality is found in the Western Ngumpin languages: Ngardi, as in (30), Jaru, as in (31), and Walmajarri, as in (32). In these languages, locative case-marked arguments of 'fear' predicates are one of a circumscribed set of locative-marked arguments which are available for cross-referencing (Ennever & Browne 2023).[12]

[12]This peculiarity extends beyond what one might strictly call apprehension to sources of emotion; it is found e.g. with predicates of anger and shame.

Table 6: Case marking of the complement of a ‘fear’ predicate across Ngumpin-Yapa languages.

	Language	Locative	Dative	Aversive
Yapa	Warlpiri		✓	✓
	Warlmanpa		✓	
Eastern Ngumpin	Bilinarra		✓	N/A
	Gurindji		✓	
	Mudburra		✓	✓
	Wanyjirra		✓	N/A
Western Ngumpin	Jaru	✓		
	Ngardi	✓		✓
	Walmajarri	✓		✓

(30) Ngardi (Western Ngumpin)
Rayin-karri-nya=**rlanyanta** yalu-ngka, **kurrkurr-ta.**
afraid-be-PST=**3SG.LCT** that-LOC **owl-LOC**
‘He is afraid of that one, the owl.’ [MMN: TEN1-2017_016-02: 212312_217107]

(31) Ngardi (Western Ngumpin; Tsunoda 1981: 113)
Yambagina nga=**nyanda** yuwa-marn-an **gunyar-a.**
child CAT=**3SG.LCT** afraid-talk-PRS **dog-LOC**
‘A child is afraid of a dog.’

(32) Walmajarri (Western Ngumpin; Hudson 1978: 23)
Ngalijarra-rla pa=**jarrangurla** lapa-rni rayin.
1DU.INCL-LOC MR1=**1DU.INCL.LCT** run-PST fear
‘He ran off scared for fear of us two.’

2.2.2 Dative

The dative case can be used to mark a complement of a source of fear in Warlpiri, as in (33), Warlmanpa, as in (34), Mudburra, as in (35), Bilinarra, as in (36), and Gurindji, as in (37), and Wanyjirra, as in (38) (other functions of the dative are also discussed in Section 2.2.4). In each of these languages, the dative case marked

NPs are treated as if they were arguments and are cross-referenced by a dative/oblique bound pronoun.

(33) Warlpiri (Yapa; Mary Laughren, p.c.)
Lani-jarri ka=npa=**rla** **warna-ku**=ju?
afraid-INCH.NPST PRS=2S=**3SG.DAT** **snake-DAT**=TOP
'Are you afraid of the snake?'

(34) Warlmanpa (Yapa)
Ngayu=ma=**rna-rla** lani-ja-nya ali-ku=nya **wangani-ku**.
1=TOP=1SG.S-**3SG.DAT** afraid-INCH-PRS that-DAT=FOC dog-DAT
'I'm afraid of that dog.' [wrl-20200221: DK, 36:12]

(35) Mudburra (Eastern Ngumpin)
Yali birrard ka-yini **dimana-wu** ba=**rla**, yali.
that frightened be-PST **horse-DAT** CAT=**3SG.DAT** that
'That one is scared of horses, that one.' [MSD: DOS1-2017_020-01: 1266566_1268319]

(36) Bilinarra (Eastern Ngumpin; Meakins & Nordlinger 2014: 367)
Nyawa yabagaru, wuugarra=rningan=ba=**rla** garra **warrija-wu**.
this little scared=AGAIN=EP=**3SG.DAT** be.PRS **crocodile-DAT**
'This kid is also afraid of the crocodile.'

(37) Gurindji (Eastern Ngumpin; Meakins & McConvell 2021: 516–517)
Wuukarra ngu=lu=**rla** karri-nyana **nyanawu-wu**
afraid CAT=3PL.S=3DAT be-PRS aforementioned-DAT
kuliyan-ku, nyampayirla-wu, crocodile-ku.
aggressive-DAT whatsitsname-DAT crocodile-DAT
'They're too afraid of those dangerous ones, whatsitsname, crocodiles.'

(38) Wanyjirra (Eastern Ngumpin; Senge 2015: 319)
Ngu=nda=**yanunggula** ribaying garriny-ana **gardiya-wu**.
REAL=2AUG.S=**3AUG.OBL** frighten BE-PRS **white_man-DAT**
'You are frightened of white men.'

In Gurindji Kriol – a mixed language arising from Gurindji and Kriol codeswitching (McConvell & Meakins 2005) – complements of the same 'fear' predicate found in Gurindji (i.e. *wuukarra*) select a complement headed by the dative preposition *bo*, as in (39), analogous to Gurindji using the dative case to mark complements of 'fear' (see Meakins 2008: 430–431).

(39) Gurindji Kriol (Mixed language; Eastern Ngumpin and English-lexified Creole)
Dat ting dat yapakayi ngakparn i bin wuukarra-karra **bo**
that thing that small frog 3SG.S PST afraid-CONT **DAT.PREP**
dat jangkarni-wan i bin jeya tumaji.
that big-NMLZ 3SG.S PST there because
'The small frog was afraid of the big one, because it was there.' [FSS: FM08_a077: 218275_222408]

2.2.3 Aversive

Mudburra, Warlpiri, Jaru, Walmajarri, and Ngardi can use the aversive case to mark complements of 'fear' predicates as exemplified in (40) for Mudburra and in (41) for Warlpiri. However, none of these languages exclusively require the aversive case in these constructions: each language may alternatively either use the locative or the dative. We are unaware of any meaningful distinctions attributable to the differential patterns of case marking.

(40) Mudburra (Eastern Ngumpin; Osgarby 2018: 227)
Nyamba-wirri wirlarnkarra ka-yini?
what-AVE scared be-PRS
'What is he scared of?'

(41) Warlpiri (Yapa; Laughren 2015: 4)
Kula=lpa nantuwu-rla wirriya warrka-karla, **laninji wanti-nja-kujaku**.
NEG=IPFV horse-LOC boy climb-IRR frightened **fall-INF-AVE**
'The boy can't get up on a horse, being scared, lest (he) fall off.'

Interestingly, aversive marked NPs, while occasionally co-occuring with 'fear' predicates, are never cross-referenced in the pronominal complex, unlike dative or locative marked NPs. For example, in Ngardi, it is possible to mark sources of fear in one of two ways: with the locative or the aversive. It is only the former (locative case-marked NPs) that may be cross-referenced in the bound pronoun. Thus compare (30) with (42).[13]

[13] Interestingly, an analogous form of variation in marking complements of 'fear' predicates is found in Warrongo which can either mark a complement of 'fear' in the locative, or else the 'fear' predicate can select an entire subordinate clause whose predicate will take a dedicated 'apprehensive' inflection (Tsunoda 2011: 619).

(42) Ngardi (Western Ngumpin)
Karrarda-karri-nyanta=rnalu maju **kunyarr-amarra kuliny-jamarra**.
fright-be-PRS=1PL.EXCL.S indeed **dog-AVE**

aggressive-AVE
'We are afraid of that vicious dog.' [DMN: LC42a: 1262081_1266907]

2.2.4 Summary of idiosyncratic case marking

In this section, we have seen that predicates of 'fear' in different Ngumpin-Yapa languages select complements in dative, locative or aversive cases. 'Fear' predicates which select for locative or aversive case fall into a restricted set of predicates which select for non-core case (i.e. case other than absolutive, ergative, or dative). It is also surprising to see such considerable variation across closely related languages.

Nevertheless, when one considers the wider polyfunctionality of these case forms, these idiosyncrasies fall into wider patterns of marking of certain semi-transitive predicates. In the case of datives, it has been noted that various emotion predicates tend to select dative objects. The range of predicates cover the semantic space of: 'afraid of', 'sulk over', 'angry with', 'happy for', 'worry about' (Wanyjirra: Senge 2015: 571). Datives also encode topics of speech activities and certain cognitive processes (e.g. 'talk about', 'forget', 'laugh at'). Therefore, while the widely attested functions of the dative appear to be generally more centered around benefactives or affected participants, the object marking of 'fear' predicates seems to fit more straightforwardly into a category of marking topics of cognition or emotional states.

Locative case marking of "objects" of semi-transitives is somewhat less reported in Ngumpin-Yapa languages than dative marking. However, in at least Jaru, Walmajarri and Ngardi, a small number of emotion predicates appear to subcategorise locative objects, including 'fear', 'shame' and 'anger' predicates (Tsunoda 1981: 58; Ennever 2021: 529). The use of locative case to mark the conceptual source of these emotional states perhaps has its origin in the marking of circumstance (i.e. 'I am afraid in the presence of *x*'). The marking of circumstance, be it a temporal or spatial location, is a more general semantic property of the locative case and is also extended somewhat transparently to the subordinating functions of the locative which (in some Ngumpin-Yapa languages) marks simultaneous tense (i.e. circumstances temporally coincident with the event depicted by the main clause).

Finally, we saw that some Ngumpin-Yapa languages exhibit aversive case marking of complements of ‘fear’ predicates. That is, a ‘fear’ predicate (which encodes apprehension) takes an aversive-marked nominal as its complement (where the aversive case also encodes apprehension independently). Lichtenberk (2016) shows that this is a typologically common construction to encode apprehension.

2.3 Auxiliaries

In addition to the encoding of apprehension via case suffixes (Section 2.1), all Ngumpin-Yapa languages possess an auxiliary that occurs in apprehensional constructions.[14] These auxiliaries are variously analysed as modal particles or complementisers and exhibit co-occurrence restrictions with certain TAM inflections. The auxiliaries which are used in finite apprehensional constructions are catalogued in Table 7. With the exception of *kajika* (Warlpiri, marks general irrealis), each are analysed in their respective grammar as an apprehensional marker (most commonly glossed ‘admonitive’). However, as we discuss in Section 2.3.2, the situation may not be this straightforward.

Table 7: Auxiliaries which can signal apprehension in Ngumpin-Yapa languages.

	Language	Auxiliary(ies)
Yapa	Warlpiri	*kalaka*
		kajika
	Warlmanpa	*kala*
		kalaka
		nga=…=nga
Ngumpin	Bilinarra	*ngaja*
	Gurindji	*ngaja*
	Mudburra	*bi(ya)*
	Wanyjirra	*ngarra*
	Ngardi	*ngarra*
	Walmajarri	*nga-rta*
		pa-rta

[14] In some languages, the auxiliary-like morpheme may be analysed as a particle. Across Ngumpin-Yapa languages, it is often difficult to distinguish the two parts of speech.

Taking Ngardi as an example, the subordinating aversive case *-Camarra* in (43) can be contrasted with the auxiliary *ngarra* in (44). In the former, the apprehension clause is non-finite, and so each constituent in the non-finite clause takes *-ngkamarra*; and in the latter, the apprehension clause is finite, and so is headed by the auxiliary *ngarra*. In both examples a preemptive clause expresses the action to be taken to avoid the undesirable outcome. The status of the preemptive clause in examples such as (44) as matrix clause or in a paratactic relationship with an apprehensional finite clause will be discussed in Section 2.3.1.

(43) Ngardi (Western Ngumpin)
Ya-nanta ma=rna kayirra-purda, **nya-ngu-ngkamarra**
go-PRS TOP.AUX=1SG.S north-ALL **see-INF-AVE**
ngaringka-rlamarra.
woman-AVE
'I am going north, for fear of the woman seeing (me).' [PSM: TEN1-2018_034-01: 1028530_1038329]

(44) Ngardi (Western Ngumpin)
Wurna=rna ya-nanta kayirra **ngarra**=yi ngaringka-rlu nya-nngi.
travel=1SG.S GO-PRS north **APPR**=1SG.NS woman-ERG see-IRR
'I am going north, lest that woman see me.' [PSM: TEN1-2018_034-01: 1008445_1010489]

There is variation in the languages across two dimensions: whether the auxiliaries are permissible in matrix clauses or not (which is taken as a distinction between auxiliaries and complementisers, in that by definition complementisers cannot occur in matrix clauses, and conversely auxiliaries are not permitted in non-finite subordinate clauses), and whether the apprehension component is encoded or not.[15]

In this section, we review the evidence for both their complementiser status and whether the auxiliaries in question are used exclusively in apprehensional constructions or if the auxiliary has apprehensional functions as just a subset of wider irrealis functions. Generally speaking the preponderance of evidence is for nearly all Ngumpin-Yapa languages to: i) have a main clause auxiliary rather than a *bona fide* complementiser; and ii) have non-dedicated apprehensionals that nevertheless show clear evidence of developing towards dedicated apprehensives.

[15]See McConvell (2006) for a diachronic analysis of Ngumpin-Yapa complementisers, including a brief discussion of whether some auxiliaries have undergone insubordination, being historically derived from complementisers.

2.3.1 Syntactic status: Main clause or subordinate?

As noted by Vuillermet & Schultze-Berndt (2018) "the relationship between precautioning and preemptive clauses may be one of subordination, coordination or simple juxtaposition." So far, we have observed non-finite subordinate apprehensional constructions (involving case suffixes) which were clearly subordinate in that the non-finite clauses rely on the main clause for the interpretation of their tense values and generally do not occur as independent clauses. However, in the case of finite clauses with apprehensional auxiliaries, there is no clear evidence that they are subordinate to a matrix clause.

Clauses with the apprehensional auxiliaries do not rely on any main clause for the interpretation of their tense values or any control properties, nor do they exhibit different morphosyntax depending on whether they occur on their own or accompanied by a preemptive clause. For example, Warlmanpa uses *nga=...=nga* regardless of whether there is a preemptive clause, as in (45), or not, as in (46). In both cases, the clause is finite and marked by the same auxiliary form.

(45) Warlmanpa (Yapa)
Ngurra-ka=rna pina pana, kula **nga**=ju=**nga** yarnunju
camp-ALL=1SG.S return GO.POT NEG **APPR**=1SG.NS=**APPR** food
ji-nnya.
burn-PRS
'I should return home, lest he not cook food for me.' [N_D03-007868, BN, 36:57]

(46) Warlmanpa (Yapa)
Ngarrka-ngu **nga**=ngu=**nga** karta pi-nya
man-ERG **APPR**=2SG.NS=**APPR** spear act_on-POT
'Careful, the man might spear you.' [H_K01-505B, DG, 19:51]

Similarly in Mudburra, the same modal auxiliaries occur with and without an additional clause expressing the preemptive event, as in (47) and (48) respectively. In (47), as with the Warlmanpa examples above, both clauses have their own set of argument relations expressed in the pronominal complex and there are no constraints on co-reference properties between the two clauses.

(47) Mudburra (Eastern Ngumpin; McConvell 1980: 72)
Karu=ma=yina-ngu-lu warra nya-ngka,
child=TOP=3PL.NS-LINK-[3]PL.S care see-IMP

bi=yina-ngu-lu bi-rnarra kaya-li=ma!
APPR=3PL.NS-LINK-[3]PL.S bite-SPEC ghost-ERG=TOP
'Take care of the children, in case ghosts bite them!'

(48) Mudburra (Eastern Ngumpin; Osgarby 2018: 59)
Bu=ngku biya-la warlaku-lu.
APPR=2SG.NS bite-SBJV dog-ERG
'The dog might bite you.'

It is therefore necessary to propose that the syntactic relationship between preemptive clauses and apprehensive clauses is simply juxtaposition and the relationship between the two clauses is likely at a discourse level, rather than a syntactic dependency.

2.3.2 Functional status: Dedicated apprehensional?

The Ngumpin-Yapa languages vary in the extent to which the auxiliary used in apprehensional utterances can be argued to be a dedicated marker of apprehension. Mudburra *bi* and Warlmanpa *nga=…=nga* are both analysed as dedicated apprehensionals, as will be discussed below. The form *ngaja* in Bilinarra and Gurindji is also analysed as a dedicated apprehensional (Meakins & Nordlinger 2014: 419); however evidence from at least Gurindji suggests that it is possibly better treated alongside *ngarra*/*ngarda* ('hypothetical') in Wanyjirra, Jaru, Walmajarri and Ngardi, rather than as a dedicated marker of apprehension.[16] Each of these will be discussed in turn.

2.3.2.1 Mudburra

In Mudburra, the auxiliary base *bi* (allomorphs: *bi*, *bu*, and *biya*) is dedicated to encoding apprehensive situations (Osgarby 2018: 104), irrespective of whether it occurs with a preemptive clause, as in (49), or without one, as in (50). This auxiliary base can be contrasted with 'hypothetical' *nya-* in (51) which is used for possible events that are not negatively evaluated.

(49) Mudburra (Eastern Ngumpin; Osgarby 2018: 178)
Biya wurruja ya-yinarra, lurrbu=rni yuwa-rra!
APPR dry become-SPC return=RSTR put-IMP
'Go and put it back, in case it dries out!'

[16] Meakins & Nordlinger (2014: 305) provide the following description of the function of *ngaja*: "(it) usually refers to something potentially dangerous occurring in the immediate discourse context."

(50) Mudburra (Eastern Ngumpin; Osgarby 2018: 105)
Biya wari-li bi-rnarra.
APPR snake-ERG bite-SPEC
'The snake might bite her/him/it.'

When used in a conditional protasis clause, the condition is negatively evaluated, as in (51).

(51) Mudburra (Eastern Ngumpin; Osgarby 2018: 107)
[Kawarraj **bi**=n=**baa** warnd-arda,]$_{\text{CONDITION}}$
forget APPR=2SG.S=COND get-SBJV
[nya=n durd ma-rrarnda wumangku=ma.]$_{\text{RESULT}}$
HYP=2SG.S hold do-SBJV dreaming=TOP
'**If** you **were in danger of** forgetting them, you would have to hold on to your totemic designs.'

2.3.2.2 Warlmanpa

Warlmanpa's apprehensional auxiliary *nga=...=nga* could be decomposed into two independent auxiliaries. The base *nga=* encodes a situation with future time reference, as in (52), often with a 'result' meaning, as in (53), when juxtaposed alongside another clause.[17]

(52) Warlmanpa (Yapa)
Nga=rna=ngu=lu turru ma-nji-nmi jaru.
FUT=1EXCL.S=2SG.NS=PL.S tell do-INC-FUT message
'We're going to start telling a story for you.' [N_D29-023128, MFN, 01:15]

(53) Warlmanpa (Yapa)
Pingka wangka, **nga**=rna purtu-ka-mi
slowly speak.IMP FUT=1SG.S hear-be-FUT
'Speak slowly, **so** I can hear you.' [H_K01-506A, DG, 32:04]

The auxiliary clitic *=nga* encodes epistemic possibility as illustrated by (54)–(55):

[17] Interestingly, Warlmanpa has a future tense inflection which does not require the use of the auxiliary base *nga=*, however *nga=* 'FUT' is only used in combination with the future inflection. It is not clear whether the auxiliary base *nga=* in these contexts is contributing further information, or has become a default auxiliary base.

(54) Warlmanpa (Yapa)
Witawita=rnalu=**nga** ngayu, but ngayu-jarra=ma=ja, little bit wiri,
small=1EXCL.S=**POT** 1 but 1-DU=TOP=1DU.EXCL little bit big
yumpa=ma wita one.
this=FOC small one
'Maybe we were all small, but us two were a bit bigger, this one was small.' [N_D29-023128, MFN, 05:34]

(55) Warlmanpa (Yapa)
Maliki yimpa=ma milpayi, kula=jana=**nga** nya-nganya.
dog this=TOP blind NEG=3PL.NS=**POT** see-PRS
'This dog is blind, he probably can't see them.' [N_D03-007868: BN: 36:01]

In Warlmanpa, an apprehensional construction is formed via the use of these two morphemes in combination with a future/potential verb inflection. Synchronically, this could be captured in a compositional analysis of *nga=* 'FUT' and *=nga* 'POT', as apprehensional situations typically occur posterior to reference time, and expect an alternative course of events to avoid the situation, presupposing a potential modality (that is, the event is uncertain). However, the compositional analysis is set back by the fact that *nga=…=nga* in combination with a future or potential verb inflection invariably results in an apprehensional reading, as in (56).

(56) Warlmanpa (Yapa)
Ngurra-ka=rna pina pa-na, kula **nga**=ju=**nga** yarnunju ji-nnya.
camp-ALL=1SG.S return go-POT NEG **FUT**=1SG.NS=**POT** food burn-PRS
'I should return home, lest he not cook food for me.' [H_K01-505B, DG: 19:51]

That is, there are no noted cases of *nga=…=nga* with a potential or future verb inflection which do not force an apprehensional reading.[18] However, when combined with a counterfactual verb inflection, the clause does not involve apprehension:

(57) Warlmanpa (Yapa)

[18] Nash (1979) analyses the meaning of auxiliaries in Warlmanpa as being dependent on the verb inflection. Hence, under this analysis, *nga=…=nga* combining with most verb inflections encodes an apprehensional meaning, but with the counterfactual inflection, *nga=…=nga* is interpreted as a conditional (result).

Kari=n pirraku-ja-rrarla=ma, **nga**=rna-ngu=**nga** ngapa
COND=2SG.S thirsty-INCH-IRR=TOP **FUT**=1SG.S-2SG.NS=**POT** water
yi-**njakurla**.
give-**CFACT**
'If you were to become thirsty, I would give you water.' [H_K01 506A, DG, 07:31]

Furthermore, non-compositional modal systems are found elsewhere in Australia. For example, Caudal et al. (2019) suggest the Iwaidja and Anindilyakwa TAM paradigms are best treated as non-compositional circumfixes, which is likely the best analysis for the Warlmanpa construction. Verstraete (2005) also investigates the functions of mood having multiple exponents in a survey of Australian languages (see also Legate 2003, Iatridou 2000, Romero 2014, Caudal & Bednall 2016).[19] Interestingly, Warlmanpa also has an auxiliary *kala(ka)* 'apprehensive', which is homophonous with *kala* 'habitual' (*kala* is also analysed as a past habitual auxiliary in Warlpiri). The contrast between *kala* 'APPR' and *kala* 'HAB' in Warlmanpa is demonstrated in (58)–(59). The choice of *nga=...=nga* or *kala(ka)* is speaker-dependent, and both combine with the same range of verb inflections.

(58) Warlmanpa (Yapa; Browne 2021: 286)
Kula=jana pa-nka-pa, **kala**=ngu-lu pi-nyi.
NEG=3PL.NS go-IMP-AWAY **APPR**=2SG.NS-3PL.S bite-FUT
'Don't go near them, lest they bite you!'

(59) Warlmanpa (Yapa; Browne 2021: 287)
Kala=lu parti-nja-na work-ku, ngayu **kala**=rnalu
HAB=3PL.S leave-MOT-PST.AWAY work-DAT 1 **HAB**=1PL.EXCL.S
ka-ngu.
be-PST
'They would leave for work, and we would stay (home).'

2.3.2.3 Bilinarra and Gurindji

Bilinarra *ngaja* is also described as exclusively denoting apprehensive situations, occurring both as a particle in matrix clauses and as a complementiser heading

[19] In Wardaman for instance, a single irrealis prefix can be used to mark either deontic or apprehensive moods depending on whether it is combined with a present tense suffix or not (Merlan 1994: 179; and see Verstraete 2005: 224–225.

finite subordinate clauses (Meakins & Nordlinger 2014: 305, 419), exemplified in (60).

(60) Bilinarra (Eastern Ngumpin; Meakins & Nordlinger 2014: 305)
Gula ga-ngga bin.ga-gurra=ma nyila=ma garu=ma, **ngaja** nyiny
NEG take-IMP river-ALL=TOP that=TOP child=TOP **APPR** drown
ya-ngu.
go-POT
'Don't take the kid to the river in case he drowns.'

Likewise, in the majority of textual examples available for Gurindji, *ngaja* denotes apprehensive situations (McConvell 1996a: 94–95; Meakins & McConvell 2021: 548–550). There are just a few examples, such as (61), however, that suggest it may have a more general function as a marker of epistemic possibility similar to the *ngarra* auxiliary found in Wanyjirra, Jaru, Walmajarri and Ngardi.

(61) Gurindji (Eastern Ngumpin; Meakins & McConvell 2021: 206)
Nyila=ma walaparr na ma-ni. **Ngaja**=nga=rla kayikayi pa-nana
that-TOP jump now do-PST **maybe**=POT=3SG.DAT chase HIT-PRS
maitbi pungayarri-lu-wayin.
maybe bull-ERG-including
'It made [the horse] jump up suddenly. Maybe a bull started chasing it.'

2.3.2.4 Wanyjirra, Jaru, Walmajarri and Ngardi

While Mudburra *bi*, Bilinarra *ngaja*, and Warlmanpa *nga=…=nga* are analysed as being fully dedicated to apprehensional situations (or, at least there is a part of the construction which unambiguously conveys apprehensional situations), the other auxiliary bases in Ngumpin-Yapa languages have either a more general irrealis meaning, or a small set of meanings associated with epistemic uncertainty which include apprehension.[20]

In Ngardi the auxiliary base *ngarda* can be used in apprehensional circumstances, as in (62), but has a more general irrealis meaning, exemplified in (63). In Wanyjirra (Senge 2015: 643), *ngarra* – in addition to more frequent admonitive uses – can sometimes simply convey epistemic possibility, "irrespective of whether it refers to a pleasant or unpleasant event" – as for example in (64) where the event is in fact favourable for the participants. Similarly, Tsunoda (1981: 204) states that *ngarra* 'possibly' (denoting epistemic possibility) in Jaru often implies

[20]For comparability, we will gloss each of these auxiliaries as 'APPR'

that the event will have an unpleasant context, but in a small number of other uses it has also been found to encode capability and is glossed as ‘can’, as in (65).

(62) Ngardi (Western Ngumpin)
Wakurra=n nga-lku puka, **ngarda**=n nyurnu-wanti-ju.
NEG=2SG.S EAT-POT rotten **HYP**=2SG.S sick-FALL-POT
‘Don’t eat the rotten (food), you might fall ill.’ [MAM: TEN1-2016_001-01: 329550_346102]

(63) Ngardi (Western Ngumpin)
Ngarda=ngala ngari ka-ngi.
APPR=1PL.EXCL.NS belongings carry-IRR
‘I can carry the stuff for us.’ [MDN: LC22b: 831428_835666]

(64) Wanyjirra (Eastern Ngumpin; Senge 2015: 643)
Nyannga maarn yan-gu marri, gangirriny **ngarra**=ngaliwa
COND cloud.ABS go-FUT off sun.ABS **APPR**=1AUG.INCL.NS
garru-wu.
stay-FUT
‘When the clouds go away, there will be sun for us.’

(65) Jaru (Western Ngumpin; Tsunoda 1981: 165)
Ngaju-nggu **ngarra**=rna bali=wu-nggu guyu nyangga langga-muwa
1SG-ERG **APPR**=1SG.S find=hit-PURP game if head-ONLY
bid-nyin-ang-gu.
stick-sit-CONT-PURP
‘I can find game if only a head is sticking out (of grass).’

Notably, however, these languages all have other auxiliary bases or particles with epistemic possibility, in addition. For example, Bilinarra has a particle *gudigada*, glossed as ‘maybe’, as in (66).

(66) Bilinarra (Eastern Ngumpin; Meakins & Nordlinger 2014: 409)
Gudigada jubu=lu=nga ya-ni-rni jarragab.
maybe just=3AUG.S=DUB GO-PST-HITH talk
‘Maybe they just came to talk.’

2.3.2.5 Warlpiri

In Warlpiri, the cognate *ngarra* can be used to indicate future tense or past irrealis, depending on the verb inflection it co-occurs with (Laughren et al. 2007: 788).

With the non-past inflection, it has a future meaning, as in (67), and with the irrealis inflection, it indicates a situation with past time reference that did not eventuate, as in (68). In either case, the situation may or may not be negatively evaluated, but the form does not mark apprehension.

(67) Warlpiri (Yapa; Laughren et al. 2007: 78)
Ngaka **ngarra**=rna-jana ma-**ni**-rra jukurra-rlu.
soon IRR=1SG.S-3PL.NS get-NPST-AWAY tomorrow-ERG
'I will go and get them tomorrow.'

(68) Warlpiri (Yapa; Laughren et al. 2007: 788)
Nama=ju langa-ngka yuka-ja **ngarra**=rna pali-**yarla**.
ant=1SG.NS ear-LOC enter-NPST IRR=1SG.S die-IRR
'An ant got into my ear and I almost died.'

This inter-language variation may have a historical explanation. Bybee et al. (1994: 211) suggest that:

> "A rather unexpected set of examples of a shift to a speaker-oriented modality is found in the development of an admonitive mood out of a marker of possibility. In most of these cases, the sense of possibility (either root or epistemic) carries with it the notion that the possible situation is unpleasant, dangerous or deleterious in some way" (see also Pakendorf & Schalley 2007).

If the *ngaCa* forms are reflexes of a single protoform, then the protoform was likely a general irrealis marker, and these reflexes synchronically may be at various stages of a shift towards dedicated apprehensional markers (i.e. apprehension may be an "invited inference" of varying strength, in terms of Traugott & Dasher 2001: 34–40).

Furthermore, Warlpiri also exhibits variation between *kalaka* and *kajika*, both of which can be used in apprehensional circumstances. In most varieties of Warlpiri, *kalaka* is a dedicated apprehensional and *kajika* marks a potential event, as in (69), though it can also be used for apprehension, as in (70).

(69) Warlpiri (Yapa; Laughren et al. 2007: 115)
Jaja-nyanu-rlu **kajika**=rla manyu wangka-mi jiliwirri
MM-ANAPH.PROP-ERG POT=3SG.DAT fun speak-NPST funny
mirntirdi-nyanu-ku wita-ku: [...]
DD-ANAPH.PROP-ERG child-DAT
'A child's maternal grandmother might say jokingly to her little grandchild: [...]'

(70) Warlpiri (Yapa; Simpson 1991: 388)

Kula=lpa=npa=jana paka-karla — lawa. **Kajika**=npa marlaja-wanti
NEG=IPFV=2SG.S=3PL.NS hit-IRR — no **POT**=2SG.S cause-fall.NPST
tarnnga, pali-mi **kajika**=npa
always die-NPST **POT**-2SG.S

'You cannot hit them. You might fall for good because of it, you might die.'

Laughren et al. (2007: 231) note that some southern Warlpiri speakers use *kalaka* to indicate the apodosis clause of conditional constructions, and as such, is not a dedicated apprehensional in these languages, further highlighting the apparent instability of apprehensional markers in Ngumpin-Yapa languages.

Table 8 summarises the extent to which auxiliaries can be seen as dedicated apprehension markers in Ngumpin-Yapa languages. The rightmost form(s) for a given language can be seen as those which are most strongly associated with apprehension. Forms such as *nga=...=nga* in Warlmanpa (in combination with future or potential inflections) encode an apprehensional reading without any other (known) possible reading. In Jaru, *ngarra* can be used to encode either apprehension, or ability, whereas *warri* is analysed as a broad irrealis marker not necessarily associated with apprehension. Whereas *ngarra* in Warlpiri, Warlmanpa, Wanyjirra, and Walmajarri can be used to indicate apprehension, the same morpheme in Jaru must be analysed as a broader irrealis marker.

In addition to this semantic categorisation of auxiliaries, there is an interesting correlation between the semantic category and its corresponding syntactic status. The auxiliaries denoting epistemic uncertainty (i.e. a broad irrealis meaning) which do not invite an apprehensional reading are also not associated with juxtaposition of another clause. This is the case for *ngarra* in Warlpiri and Warlmanpa (in Warlpiri, *ngarra* specifically has future temporal reference), which are indicated by diamonds in Figure 5. The next semantic categorisation presented above in Table 8 is implied apprehension, meaning that the auxiliary can signal epistemic uncertainty, but is more often associated specifically with apprehension – these auxiliaries are frequently (but not necessarily) juxtaposed with a preemptive clause.[21] This is the case for *ngarda* in Ngardi and *ngarra* in Wanyjirra, as represented by the circles in Figure 5.

The next logical syntactic status along this cline is for the auxiliary to only occur in subordinate clauses (due to the high frequency of being juxtaposed with

[21]Given the syntactic status of adjoined clauses in Ngumpin-Yapa languages, and Australian languages more generally (Nordlinger 2006, Hale 1976), we use the term *juxtaposed* to mean that a clause is frequently found preceding/following another clause.

Table 8: Categorisation of irrealis auxiliaries in Ngumpin-Yapa languages by their association with apprehensional readings. Different kinds of underlining indicate cognate forms.

		Apprehension			
		None	Implied		Encoded
	Invites apprehensional reading	N	Y	Y	Y
	Other possible reading(s)	Y	Y	Y	N
	Broad irrealis meaning	Y	Y	N	N
Yapa	Warlpiri	*ngarra, marda*		*kajika kalaka*[a]	
	Warlmanpa	*marta*	*ngarra*		*nga=…=nga kala(ka)*
Eastern Ngumpin	Bilinarra	*gudigada*			*ngaja*
	Gurindji			*ngaja*	
	Mudburra	*nya*	*barra*		*bi*
	Wanyjirra	*wayi, marri*	*ngarra*		
Western Ngumpin	Jaru	*warri*		*ngarra*	
	Ngardi	*parda, mayi*	*ngarda*		
	Walmajarri	*kuyungurla*	*ngarta, parta*		

[a]As noted earlier, some speakers of Warlpiri alternatively use *kalaka* in conditional constructions without apprehension.

a preemptive clause). Interestingly, no auxiliaries in Ngumpin-Yapa languages have reached this stage, although this can be seen in other Australian languages, such as *karningka* in Jingulu which is a dedicated apprehensional complementiser (see Section 1.1) – represented by the squares. While this syntactic status is not instantiated in Ngumpin-Yapa languages, the semantic function is: *ngaja* in Bilinarra and *bi* in Mudburra, for example, are analysed as dedicated apprehensional markers, however they can still surface without an additional juxtaposed clause, suggesting they are not bona fide complementisers (represented with triangles). Thus, while there is a clear correlation between syntactic status and semantic function, this is not a perfect correlation. Moreover, the correlation suggests a diachronic path towards these auxiliaries and particles becoming complementisers on the one hand, and dedicated markers of apprehension on the other.

Finally, there are some cases where both a dedicated case morpheme and a non-dedicated auxiliary construction can co-occur as in (71). In these cases, it seems the apprehensive clause expands upon the negative event associated with the referent – here the addressees are being taken away for fear of the man, which is a nominal taking the aversive case, followed by a clause which expands upon the negative event associated with the men.

(71) Ngardi (Western Ngumpin)
Ka-ngku=rna=nyurra **ngantany-jamarra** **ngarda**=lu=nganpa
carry-POT=1SG.S=2PL.NS **man-AVE** **APPR**=3PL.S=1PL.INCL.NS
luwa-rnngi.
spear-IRR
'I will take all of you for fear of the men, lest they (the men) spear us.'
[MRN: LC35b: 322091_328446]

3 Discourse functions

To conclude this paper, we now discuss the discourse functions (speech acts) instantiated by the different apprehensional constructions we have presented. Apprehensional constructions (i.e. clauses involving apprehensional morphemes, with or without an overt preemptive clause) in Ngumpin-Yapa are frequently associated with two kinds of discourse functions: directives (Section 3.1) and assertives (Section 3.2). Apprehensional constructions in directive function compel an addressee to some action and entail a negative outcome should the addressee not comply. Apprehensional constructions in assertive function on the other

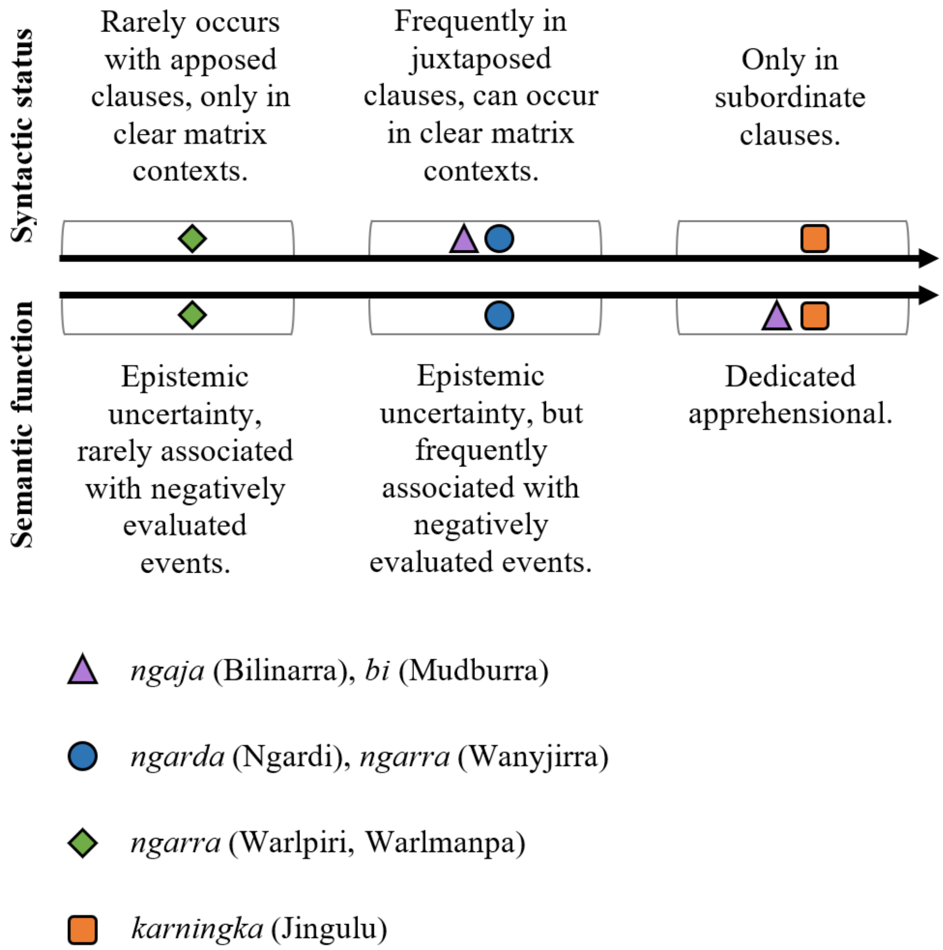

Figure 5: Correspondences between syntactic status and semantic function of auxiliaries/particles.

hand are a type of representative speech act that motivates a course of events, again by expression of a less desirable course of events. The primary morphosyntactic correspondences are that apprehensional directives make use of irrealis modalities often with second person subjects (implied or overt), whereas apprehensional assertives make use of realis modalities and describe events which affect participants with any personhood but most frequently first persons – summarised in Table 9.

Table 9: Typical modality and person values for the types of apprehensional constructions in Ngumpin-Yapa languages.

Speech act type	Typical modality	Typical affected participant
Directive	Irrealis	2nd (1st person occasionally)
Assertives	Realis	Any (1st person most common)

3.1 Directives

Apprehensional constructions are often associated with directives, either instructing (commands) an addressee to perform some action (or avoid performing some action) in order to avoid an undesirable outcome, or suggesting (entreating) an action to this effect. There are three main morphosyntactic strategies for this. The first two involve either a direct command (imperative clause) or indirect command (potential/irrealis clause) denoting the instructions and an associated apprehensional construction (either a full clause, or a nominal with relational function marked for apprehension). The third is a conditional preemptive clause which signals an event which will lead to an undesired event. These strategies are summarised in Table 10, and discussed in more detail below. Interestingly, while the first two types of preemptive clauses can be negated with an associated change of illocutionary meaning ('do/do not do a certain action'), the conditional clause strategy is inherently intended not to be heeded in its positive polarity form (i.e. do not perform this action) even without the presence of any clausal negation.

3.1.1 Direct command

An apprehensional construction may involve an overt preemptive clause which serves as a command; its morphosyntactic realisation involves a verb in the imperative inflection. These constructions convey that the addressee is expected (i.e. deontic modality) to perform the action denoted by the imperative clause, which will then avoid the undesired entity (for a case marked nominal in an aversive function, as in (72)), or avoid an undesired event, as in (73), where the clause containing the predicate is headed by the auxiliary *kala* 'HAB'.

Table 10: Summary of directives involving apprehension.

Speech act	Morphosyntax	Expected response
Direct command	Imperative construction and either (i) aversive dependent; or (ii) associated finite apprehensional clause.	Addressee is expected to perform this action to avoid something undesired.
Indirect command	Irrealis verbal inflection and either (i) aversive dependent; or (ii) associated finite apprehensional clause.	Addressee is encouraged to perform this action to avoid something undesired.
Conditional warning	Conditional complementiser with finite matrix apprehensional clause.	Addressee should avoid allowing some action occurring to avoid something undesired.

1. Aversive dependent

(72) Jaru (Western Ngumpin; Tsunoda 1981: 58)
Wanyja-rra gardiya-rla!
leave-IMP white_man-LOC
'Leave it for fear of the white man!'

2. Finite apprehensional clause

(73) Warlmanpa (Yapa)
Lamarta-ka jawurra=nya, kala=nga ngapa walikwalik-wa-nmi.
hold-IMP cup=FOC APPR=POT water spill-fall-FUT
'Hold the cup, lest water spill.' [wrl-20190205, DK, 34:01]

3.1.2 Indirect command

Apprehensional constructions may also occur with main clauses that comprise a verb in an irrealis mood and take the form of indirect commands. In these cases, a preemptive clause is realised with irrealis modality (typically through an irrealis inflection, though this is language-dependent), with either an aversive-marked nominal (or non-finite subordinate clause), as in (74), or with a finite clause with an apprehensional auxiliary base, as in (75) and (76). Unlike imperatives, the addressee is not deontically *expected* to perform the action referred to by the irrealis clause, but the speaker believes it would be beneficial (typically for the addressee) to do so.

(74) Walmajarri (Western Ngumpin; Richards & Hudson 2012)
Ngajirta nga-n jatpara-karra yan-ta **purangu-rlamarra**.
NEG MR2-2SG.S stop-MANN go-IRR sun-AVE
'Don't keep stopping, go, for fear of the sun (getting too hot).'

(75) Ngardi (Western Ngumpin)
Wanja-ku=n **ngarda**=ngkurla wanti-ngi jina-ngka.
leave-POT=2SG.S **APPR**=2SG.LCT fall-IRR foot-LOC
'You should leave it (the axe) lest it fall on your foot.' [PSM: TEN1-2020_004-01: 26146_37777]

(76) Ngardi (Western Ngumpin)
Wakurra=n walya-ngka yirra-ku yawiyi, **ngarda** wanti-ngi.
NEG=2SG.S ground-LOC place-POT poor_thing **APPR** fall-IRR
'You shouldn't put (the kid) down on the ground, he might fall over.' [PSM: TEN1-2020_006-01: 245249_252587]

Finally, there are apprehensional constructions where the preemptive clauses are protases of conditionals. In these cases, the matrix clause is marked with the apprehensive auxiliary, and a finite subordinate clause, headed by a conditional, signals an event which will lead to the apprehensive situation eventuating. That is, if the event depicted by the protasis clause becomes true, the undesired event will occur. The inferred directive of these constructions is to *prevent* the event of the conditional clause from unfolding. Unlike the commands detailed above, the conditional warning has a purely inferred directive component. Furthermore, conditional warnings must have a finite apprehensional component (i.e. using the apprehensional auxiliary). These can either serve as warnings in specific circumstances, as in (77), or more general warnings about the world, as in (78).

(77) Warlmanpa (Yapa)
Kari=n kuyu nga-rninjakurla, **nga**=n=**nga**
COND=2SG.S meat eat-CFACT APPR=2SG.S=APPR
murrumurru-ja-ma.
sick-INCH-POT
'If you were to eat this meat, you might become sick (i.e. don't eat the meat).' [H_K01 506A, DG, 04:42]

(78) Warlpiri (Yapa; Laughren et al. 2007: 518)
Nyunypa-ngku ka luwa-rni kiwinyi=ji kuyu nyanungu-ku. Kala
saliva-ERG PRS shoot-NPST mosquito=TOP meat that_one-DAT but
kaji=lpa yapa luwa-karla warlura-rlu=ju, ngula=ju **kalaka** milpa
COND=IPFV person shoot-IRR gecko-ERG=TOP that=TOP APPR eye
karltara-jarri-mi.
blind-INCH-NPST
'It spits on mosquitoes which it catches thus to eat. If the gecko hits a person (with its spittle), then the person can go temporarily blind in the eyes.'

Some Ngumpin languages do not have this conditional apprehensional construction: for example, in Wanyjirra and Ngardi, for a combination of a conditional subordinate clause with a *ngarda/ngarra* marked main clause, the main clause denotes only possibility, and is rarely associated with apprehension, as exemplified in (79)–(80).

(79) Ngardi (Western Ngumpin)
Kaji=rna kuyi ka-ngku-rni kupa-ku **ngarra**=yi=n.
COND=1SG.S meat carry-POT-HITH cook-POT IRR=1SG.NS=2SG.S
'If I bring back this meat, you might cook it for me.' [PSM: TEN1-2020_006-01: 245249_252587]

(80) Wanyjirra (Western Ngumpin; Senge 2015: 643)
Nyanga maarn yan-gu marri, gangirriny **ngarra**=ngaliwa garru-wu.
COND cloud go-FUT off sun IRR=1AUG.INC.NS stay-FUT
'When the clouds go away, there will be the sun for us.'

Mudburra allows a similar construction, in which the protasis can be apprehensional (marked by the *bi*= auxiliary base). In this case, the condition is something to be avoided, and the main clause signals an action to be undertaken to revert the undesirable event. This is exemplified in (81), where the condition is the addressee forgetting a referent (totemic designs), which is undesired, and the main clause is that the addressee would have to be sure to hold on to the totemic designs. By performing the main clause event under the appropriate conditions (undesired conditions), the addressee reverts to a state where the undesired conditions are no longer met.

(81) Mudburra (Eastern Ngumpin; Osgarby 2018: 107)
[Kawarraj **bi**=n=**baa** warnd-arda,]$_{\text{CONDITION}}$
forget **appr**=2sg.s=**cond** get-sbjv
[nya=n durd ma-rrarnda wumangku=ma.]$_{\text{RESULT}}$
appr=2sg.s hold do-sbjv dreaming=top
'If you were in danger of forgetting them, you would have to hold on to your totemic designs.'

3.2 Assertives

Apprehensional constructions in Ngumpin-Yapa languages are also used as assertives, in particular providing the reasoning behind a speaker's actions. Examples are given in (82)–(85) which demonstrate the wide range of temporal references permitted in the preemptive clause: the example in (82) has present reference; (83) has past reference; and finally (84)–(86) have future reference. The apprehensional component, either a finite clause with an apprehensive auxiliary and irrealis modality, or a nominal with aversive case marking (or a non-finite subordinate clause with aversive case marking), obtains its temporal reference from the preemptive clause.

(82) Ngardi (Western Ngumpin)
Ngu=rna jaku=ya-nanta, **ngarda**=yi=lu paja-rnngi kunyarr-u.
seq=1sg.s depart=go-prs **appr**=1sg.ns=3pl.s bite-irr dog-erg
'I'm getting out of here, **lest** those dogs bite me!' [PSM: TEN1-2017_010-01: 954694_959466]

(83) Ngardi (Western Ngumpin)
Ngu=lu=yanu wiriny=ma-ni yard-ta=wali ngantany jayin-kurlu
seq=3pl.s=3pl.ns bind=get-pst yard-loc=then man chains-prop
jina **ngarda**=lu yaparti-ngi.
foot **appr**=3pl.s run-irr
'They tied the aboriginal men up in the yard (by their) feet with chains, lest they flee.' [PSM: TEN1-2017_007-02: 72766_82623]

(84) Ngardi (Western Ngumpin)
Ya-nku=rna yalu-rlamarra, **ngarda**=yirla wangka-nyanku.
go-POT=1SG.S that-AVE **APPR**=1SG.LCT speak-IPFV.POT
'I will go on account of that one, lest she speak with me.' [MMN: TEN1-2017_026-01: 1168486 _1207635]

(85) Warlmanpa (Yapa)
Ngapa-kuma=rna mulpu-nga purluny-ya-n.
water-AVE=1SG.S shelter-LOC enter-go-FUT
'I will enter shelter for fear of the rain.' [H_K01 506A, DG, 39:05]

Apprehensional constructions can be used not in reference to a specific event but rather to justify claims about generally observed behaviours or phenomena. In (86), a speaker claims that cars cannot go over sandhills, the justification being that the car will get bogged, which is marked with apprehensional morphology.

(86) Warlpiri (Yapa; Laughren et al. 2007: 116)
Jilja ka pirli=piya parntarri-nja-ya-ni kula=lpa murduka=yi
sandhill PRS rock=SEMBL crouch-INF-go-NPST NEG=IPFV car=CONT
jilja-ngka ya-ntarla, lawa – **kalaka** yuka-mi.
sandhill-LOC go-IRR NEG **APPR** sink-NPST
'A sandhill stretches out like a rocky mesa—a car can't go over a sandhill, as it's likely to get bogged.'

4 Conclusion

This chapter has provided the first in-depth analysis of apprehensional constructions of Ngumpin-Yapa languages. These languages exhibit a case suffix which can mark a nominal denoting a feared referent (or a feared event metonymically associated with a referent), or it can mark a non-finite subordinate clause denoting a feared event. In some cases it is difficult to distinguish which particular function is instantiated in a given utterance, which is a common analytical problem in Australian languages (Hale 1982). While most Ngumpin-Yapa languages have this construction available, few use the aversive case suffix to mark the complement of a 'fear' predicate, and it is more common to mark these complements using dative or locative suffixes which still trigger verbal indexation (thus distinct from adjunct uses of the dative and locative cases). In most Ngumpin-Yapa languages, the form of the aversive suffix is an increment to another case (particularly dative, as in Warlmanpa, or locative, as in Ngardi), which can be seen in other Australian languages as well (e.g. Pintupi).

Most Ngumpin-Yapa languages also have an auxiliary which marks finite clauses as denoting undesired eventualities – while this has been found in non-Pama-Nyungan languages, this is rare for Pama-Nyungan languages. There is great interlanguage variation as to whether these auxiliaries are more like complementisers (i.e. in these languages, these constructions rely on a matrix preemptive clause), and to the extent to which these auxiliaries exclusively encode apprehension or whether other interpretations are available. Generally, these other interpretations are general irrealis meanings, which leaves available an apprehensive interpretation. This variation indicates that within these languages, apprehension is not a diachronically stable phenomenon, particularly striking in one form which appears to be a shared inherited form *ngaCa* which is dedicated to apprehensional meanings in some languages but has a more general irrealis function in others, as well as exhibiting variance regarding its syntactic status. Furthermore, the interpretation of apprehension is tied to the verb inflection, as well as the auxiliary. Finally, the chapter discussed the most common usage of apprehensional constructions: directives of varying strength, and assertives, which primarily provide the reasoning for a speaker's actions.

Abbreviations

1/2/3	1st/2nd/3rd person
A	Agent
ABL	ablative
ABS	absolutive
ACC	accusative
ACS	accessory
ADM	admonitive
ALL	allative
ANAPH.PROP	anaphoric/reflexive pronoun
ANTIC	anticipatory
APPR	apprehensive
AUG	augmented
AUX	auxiliary
AVE	aversive
CAT	catalyst
CAUS	causative
CFACT	counterfactual
COM	comitative
COND	conditional
CONT	continuous
CUST	customary aspect
DD	daughter's daughter
DAT	dative
DIST	distal demonstrative
D.LOC	derivational locative
DU	dual
DUB	dubitative
EFF	effector
ERG	ergative
EXCL	exclusive
FOC	focus
FUT	future
HAB	habitual
HITH	hither
HYP	hypothetical
IMP	imperative
INCL	inclusive

INCH	inchoative	OBSC	obscured
INF	infinitive	PASSP	passive perfective
INST	instrumental	PERF	perfect
IPFV	imperfective	PL	plural
LEST	'lest'; apprehensional marker	POT	potential
IRR	irrealis	PREP	preparative
LCT	locational	PROP	proprietive ('having')
LINK	linker	PRS	present
LOC	locative	PST	past
MANN	manner	RDP	reduplication
MIN	minimal	PURP	purposive
MM	mother's mother	REAL	realis
MOT	associated motion	REFL	reflexive
MR1/2/3	modal root 1/2/3	RSTR	restrictive
NARR	narrative	S	subject
NEG	negative	SBJV	subjunctive
NMLZ	nominalizer	SEMBL	semblative
NPST	non-past	SG	singular
NOM	nominative	SPEC	speculative
NS	non-subject	TOP	topic
O	object	VBLSR	verbaliser
OBL	oblique		

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